

Lab. 8 - Project Report

MVG and Variants Performance Analysis (Application-Specific)

Application Scenarios and Effective Priors

We consider five application scenarios defined by prior π and costs

C_{fn}, C_{fp}

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Scenario	$\pi(Prior)$	C_{fn}	C_{fp}	$EffectivePrior(\tilde{\pi} = \frac{\pi C_{fn}}{\pi C_{fn} + (1 - \pi) C_{fp}})$
1	0.5	1.0	1.0	0.5
2	0.9	1.0	1.0	0.9
3	0.1	1.0	1.0	0.1
4	0.5	1.0	9.0	0.1
5	0.5	9.0	1.0	0.9

Observation: Increasing false positive cost

C_{fp}

decreases the effective prior

$\tilde{\pi}$

which means the system becomes more conservative (more likely to classify as impostor). Vice versa for

C_{fn}

Binary Classification: MVG Performance

Normalized DCF and MinDCF Summary

$EffectivePrior(\tilde{\pi})$	DCF (Normalized)	MinDCF (Normalized)
0.5	0.511	0.506
0.9	2.046	0.827
0.1	2.211	0.907

Observation: MinDCF increases at extreme priors due to imbalance. Actual DCF is significantly higher than MinDCF, suggesting **calibration loss**.

Bayes Error Plots (Shown in Images)

- **MinDCF vs. Prior Log Odds** is smooth for all models.
- **Actual DCF** consistently lies above MinDCF, with Naive Bayes showing higher deviation — indicating it's **less calibrated**.

Model Comparison

- **MVG** generally performs best in terms of MinDCF.
- **Tied Gaussian** performs closely and can even outperform MVG when sample size is small or when covariance assumptions match.
- **Naive Bayes** shows higher MinDCF and greater deviation between actual and minimum DCF, especially for low priors, confirming poor calibration and less expressive modeling.

Conclusion: MVG is the best model in terms of accuracy (MinDCF), while Tied is a good tradeoff. Naive Bayes is simple but suffers in both discrimination and calibration.

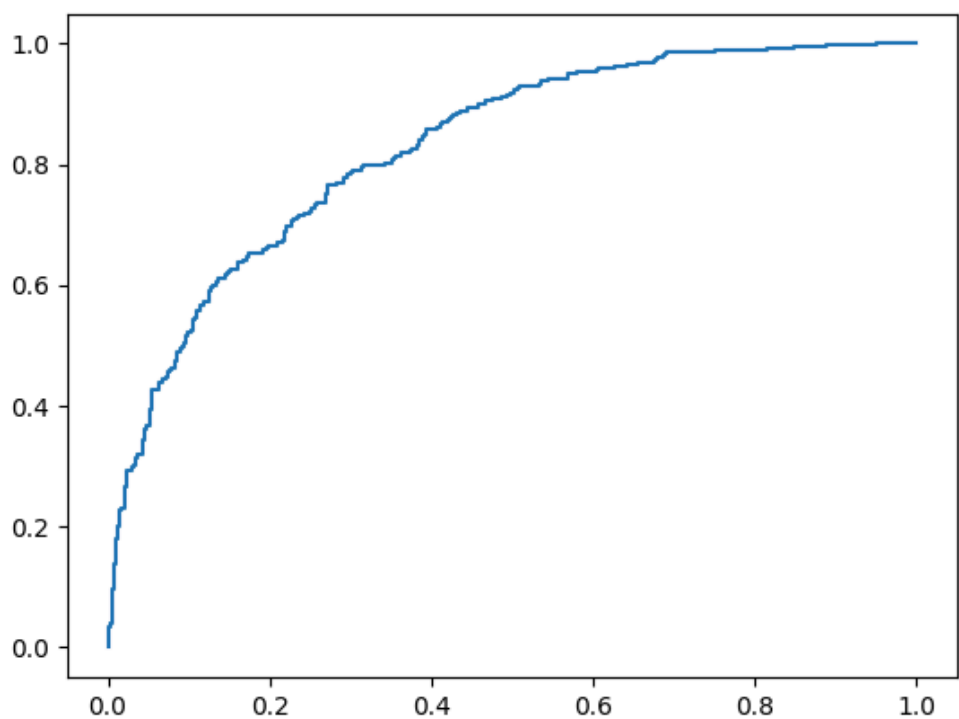
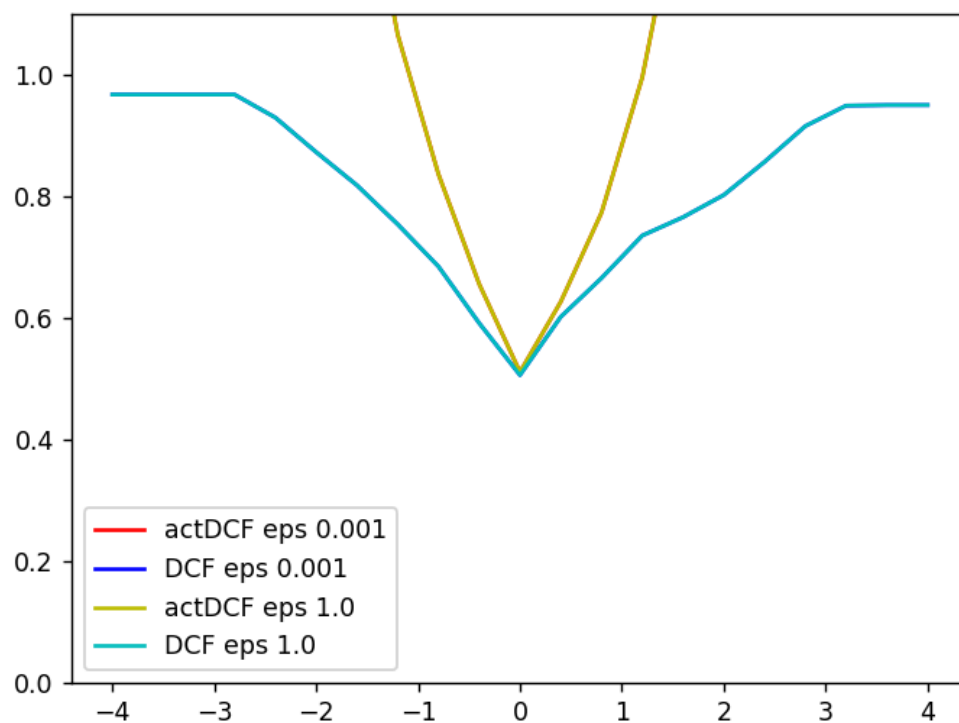
Multiclass Performance

- **Eps = 0.001:** Normalized Bayes Risk = **0.933**
- **Eps = 1.0:** Normalized Bayes Risk = **0.808**

Observation: Increasing (ϵ) (regularization) reduces Bayes risk, suggesting that the model benefits from **smoother probability estimates** in multiclass settings.

Summary

- **Model ranking is consistent:** MVG > Tied > Naive Bayes.
- **Calibration:** Actual DCF deviates from MinDCF more in Naive Bayes.
- **Application sensitivity:** Strong priors (0.1 or 0.9) and cost asymmetries heavily impact MinDCF.
- **Best tradeoff:** MVG provides reliable accuracy and calibration, while Tied can be competitive in simpler or low-data scenarios.



[[210 113 61] [137 191 111] [53 98 230]]

Binary task

Prior 0.5 - Cfn 1 - Cfp 1 [[293 96] [109 304]]

DCF (non-normalized): 0.256

DCF (non-normalized, multiclass code): 0.256

DCF (normalized): 0.511

DCF (normalized, multiclass code): 0.511

MinDCF (normalized, slow): 0.506 (@ th = -1.230986e-01)

MinDCF (normalized, fast): 0.506 (@ th = -1.230986e-01)

Prior 0.9 - Cfn 1 - Cfp 1 [[258 75 [144 325]]
DCF (non-normalized): 0.205
DCF (non-normalized, multiclass code): 0.205
DCF (normalized): 2.046
DCF (normalized, multiclass code): 2.046
MinDCF (normalized, slow): 0.827 (@ th = -1.464205e+01)
MinDCF (normalized, fast): 0.827 (@ th = -1.464205e+01)

Prior 0.1 - Cfn 1 - Cfp 1 [[318 132] [84 268]]
DCF (non-normalized): 0.221
DCF (non-normalized, multiclass code): 0.221
DCF (normalized): 2.211
DCF (normalized, multiclass code): 2.211
MinDCF (normalized, slow): 0.907 (@ th = 1.715560e+01)
MinDCF (normalized, fast): 0.907 (@ th = 1.715560e+01)

Prior 0.5 - Cfn 1 - Cfp 9 [[318 132] [84 268]]
DCF (non-normalized): 1.105
DCF (non-normalized, multiclass code): 1.105
DCF (normalized): 2.211
DCF (normalized, multiclass code): 2.211
MinDCF (normalized, slow): 0.907 (@ th = 1.715560e+01)
MinDCF (normalized, fast): 0.907 (@ th = 1.715560e+01)

Prior 0.5 - Cfn 9 - Cfp 1 [[258 75] [144 325]]
DCF (non-normalized): 1.023
DCF (non-normalized, multiclass code): 1.023
DCF (normalized): 2.046
DCF (normalized, multiclass code): 2.046
MinDCF (normalized, slow): 0.827 (@ th = -1.464205e+01)
MinDCF (normalized, fast): 0.827 (@ th = -1.464205e+01)

Multiclass task
Eps 0.001 [[205 111 56] [145 199 121] [50 92 225]]
Empirical Bayes risk: 0.560
Normalized empirical Bayes risk: 0.933

Eps 1.0 [[216 77 31] [146 236 143] [38 89 228]]
Empirical Bayes risk: 0.485
Normalized empirical Bayes risk: 0.808